

Near-perfect Clustering Based on Recursive Binary Splitting Using Max-MMD

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Abstract—We develop novel clustering algorithms for functional data when the number of clusters K is unspecified and also when it is specified. These algorithms are developed based on the Maximum Mean Discrepancy (MMD) measure between the empirical distributions associated with two sets of observations. The algorithms recursively use a binary splitting strategy to partition the dataset into two subgroups such that they are maximally separated in terms of an appropriate weighted MMD measure. When K is unspecified, the proposed clustering algorithm has an additional step to check whether a group of observations obtained by the binary splitting technique consists of observations from a single population. We also learn K directly from the data using this algorithm. When K is specified, a modification of the previous algorithm is proposed which consists of an additional step of merging subgroups which are similar in terms of the weighted MMD distance. The theoretical properties of the proposed algorithms are investigated in an oracle scenario that requires the knowledge of the empirical distributions of the observations from different populations involved. In this setting, we prove that the algorithm proposed when K is unspecified achieves perfect clustering while the algorithm proposed when K is specified has the perfect order preserving (POP) property. Extensive real and simulated data analyses using a variety of models having location difference as well as scale difference show near-perfect clustering performance of both the algorithms which improve upon the state-of-the-art clustering methods for functional data.

Index Terms—Characteristic kernel, empirical distributions, functional data, Maximum mean discrepancy, Oracle analysis, perfect order preserving, Rand index.

I. INTRODUCTION

CLUSTERING of a given dataset aims at partitioning the data into subgroups such that each subgroup is as homogeneous as possible and any two distinct subgroups are as heterogeneous as possible. With modern sensing and imaging technologies routinely capturing high-resolution trajectories and fields, the resulting (functional) data is recorded as a curve, surface,

or other continuous object rather than as a finite-dimensional vector. This poses distinct statistical challenges since one must account for the infinite-dimensional nature of the observations and the dependence structures embedded in the continuous variation. For functional data, the problem of clustering has received quite a lot of attention in the past since many of the multivariate techniques may not be directly applicable, and the fact that one could also incorporate the functional nature of the data while building a clustering procedure (e.g., also consider the derivative curves in addition to the original data curves). One of the classical approaches for clustering of functional data is to define an appropriate notion of distance between two data curves e.g., the usual L_2 distance and their modifications (see [1], [2], [3], [4], [5], [6]) followed by a k-means/k-medoids algorithm. A second approach which is typical to functional data is to use projections of the data onto a suitable finite dimensional subspace and then use an appropriate multivariate clustering technique. Functional data may be projected in a finite dimension mainly in two ways: (i) for fully observed curves, one can project the curves onto an orthonormal basis in the function space (see [7], [8], [9], [10]), and (ii) represent the discretely observed data through an appropriate truncated basis expansion using spline basis, principal component basis, wavelet basis, among others (see [11], [12], [13], [14] and [15]). While some methods for functional clustering use model-based techniques (see [16], [17], [18], [19], [20], [21]), a few papers use pseudo-density based approaches (see [22], [14], [23]). There is also a growing literature on using deep learning and auto-encoder architectures for functional clustering (see [24], [25], [26]).

Majority of the literature on functional clustering deals with the situation where there is difference in location. However, the case where the difference is in the scale, or more generally, in the distributional structure has only been sparsely addressed. Furthermore, the papers which consider scale difference mostly require additional difference in the location for the method to work (see, for example, [10]). On the other hand, model based or pseudo density based approaches (which potentially can deal with distributional differences) are too restrictive in their approaches. Model based approaches usually make Gaussian or some other parametric mixture model assumptions while pseudo-density based procedures use approximations of the likelihood due to the nonexistence of a conventional notion of density for functional data. However, the existing work in these directions are still restricted to detecting difference only in the location. Clustering methods based on modern machine learning

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techniques typically use modifications of the k-means objective function and thus fail to detect differences in the scale.

One possible approach to detect differences in location/scale/distributional structure is to consider a suitable notion of distance between two clusters or distributions and to use this distance to develop the clustering algorithm. In this paper, we develop novel nonparametric clustering algorithms whose building block is an appropriate notion of distance between two sets of observations. This is unlike most of the distance based clustering techniques which generally use distances between observations to build the clustering algorithm.

Recall that the objective of clustering is to create homogeneous groups that are heterogeneous between themselves. The distance function approach taken in this paper achieves both ends. If we start with a single observation as a group, then we can add to it those observations which maximize the distance between the group thus formed and the rest of the dataset. The choice of the distance is crucial since it must simultaneously ensure that the observations that are added to the singleton group are as “similar” to the existing observation as possible while also ensuring that the distance between the two groups is as large as possible. This paper will exploit the Maximum Mean Discrepancy (MMD) [27] as a measure of distance between two groups of observations. Such an application of MMD for functional data clustering is novel to the best of our knowledge. We will show later in the paper that the MMD ensures that both within group homogeneity and between group heterogeneity can be achieved. To provide an insight into this phenomenon, we state the following property of MMD: given two distributions P and Q , the MMD between any two mixtures $\alpha P + (1 - \alpha)Q$ and $\beta P + (1 - \beta)Q$ is strictly smaller than the MMD between P and Q with equality holding if and only if $(\alpha, \beta) = (0, 1)$ or $(1, 0)$ (cf. Remark 3). Thus, if we have a two-class clustering problem with unknown empirical distributions $\hat{P}$ and $\hat{Q}$ corresponding to the two samples, the above property ensures that if we want to maximize the MMD between any two sub-groups of this data, this is achieved when the data is partitioned “perfectly” (cf. Definition IV.I), that is, when the clusters are pure. Note that any subgroup of the original data can be viewed as a sample from a mixture distribution involving $\hat{P}$ and $\hat{Q}$, and this link will be exploited heavily in this paper to develop and investigate the clustering algorithms. The above property also ensures the following. Suppose that we start with a partition with one single observation as a group (the most obvious choice of a pure group), say C_1 , and the rest of the data as another group, say C_2 . Then, inclusion of another observation from C_2 to C_1 yielding the maximum MMD between the new groups also ensures that the transferred observation belongs to the same population as the observation already in C_1 . Thus, heterogeneity between groups and homogeneity within groups are preserved simultaneously. Since in a typical clustering problem, the knowledge of $\hat{P}$ and $\hat{Q}$ are unavailable, we may view this as an oracle version of the clustering problem. Indeed, the clustering algorithms proposed in this paper are motivated by the above crucial phenomena, and the understanding of the performance of these algorithms can be obtained by a detailed investigation in such an oracle scenario (see Section IV). Our clustering algorithms will exploit

the above phenomena to carefully find a path to achieve the optimal partition in a finite number of steps instead of searching for all possible partitions.

A. Main Contributions

The main contributions of the paper are as follows:

- 1) *A novel MMD-based clustering paradigm:* We introduce Algorithm CURBS-I (for unspecified K) and Algorithm CURBS-II (for specified K), both built on a common MMD-guided binary splitting mechanism that fundamentally differs from classical partitioning and hierarchical approaches. Algorithm CURBS-I adaptively discovers cluster structure through recursive splitting, while the Algorithm CURBS-II employs a principled split and merge strategy to obtain the prescribed number of clusters.
- 2) *Comprehensive oracle understanding and theoretical guarantees:* We develop a novel theoretical analysis of the clustering algorithms in an oracle scenario, where the empirical distributions of the populations are assumed to be known. The oracle analysis reveals that the MMD curve exhibits characteristic shapes determined only by pairwise MMD values and the number of populations K . This structural property of the MMD curve provides a geometric roadmap and motivates both the splitting algorithm BS and the single cluster check algorithm SCC. The latter avoids doing computationally intensive permutation testing by employing a carefully chosen data-driven threshold. The oracle analysis also establishes asymptotic perfection of an oracle version of the Algorithm CURBS-I and the perfect order-preserving (POP) property of an oracle version of the Algorithm CURBS-II. This theory forms the backbone of the near-perfect empirical performance of both algorithms.
- 3) *Automatic learning of the number of clusters via Algorithm CURBS-I:* Algorithm CURBS-I provides a fast and principled estimate of K without external validity measures (e.g., Silhouette index, gap statistic, elbow method), and naturally allows $\hat{K} = 1$, a possibility that is typically excluded when these measures are used.
- 4) *Ability to detect general distributional differences:* By exploiting the full metric structure of MMD and the associated RKHS embeddings, both procedures adapt to arbitrary distributional geometry and are thus able to detect general differences including location and scale (covariance structure). Consequently, they can provide accurate clustering even when populations differ solely in scale. This capability is not enjoyed by many competing methods.
- 5) *Practical, stable, and scalable algorithms:* The proposed algorithms have polynomial-time complexity (typically $O(n^2)$) with a low memory overhead, terminate in finitely many steps without requiring iterative optimization, and are completely deterministic thereby producing identical results across runs. They are robust to moderate noise and outliers and require only a single hyperparameter (kernel bandwidth), which is computed from the data.

The remaining part of this paper is organized as follows. In Section II, we give a brief introduction to the MMD measure. In Section III, we propose two clustering algorithms for the scenario where the number of clusters is unspecified (Algorithm CURBS-I) as well as when it is specified (Algorithm CURBS-II). In Section IV, we provide the oracle versions of these algorithms, referred to as Algorithms CURBS-I* and CURBS-II*, respectively, in order to theoretically investigate the behaviour in the oracle setting of the algorithms proposed in Section III. We establish that the Algorithm CURBS-I* achieves perfect clustering while the Algorithm CURBS-II* enjoys the POP property in this setting. A detailed simulation study and some real data analyses are provided in Section V and we compare the proposed algorithms with some of the existing clustering procedures for functional data. We conclude the paper with some additional remarks and future scope of research in Section VI.

II. PROLOGUE: A METRIC ON THE SPACE OF PROBABILITY DISTRIBUTIONS

Let $\mathcal{P}$ be the set of Borel probability measures on a separable metric space $\mathcal{X}$. A well known measure of discrepancy between two measures $P, Q \in \mathcal{P}$ is the Maximum Mean Discrepancy (MMD) [27] which is defined as

$$\text{MMD}(P, Q) = \sup_{f \in \mathcal{F}} \left| \int f dP - \int f dQ \right|$$

for an appropriate class of functions $\mathcal{F}$. The class $\mathcal{F}$ is chosen so that it is rich enough to discriminate between any two distinct probability measures. For instance, $\mathcal{F}$ can be chosen to be the space of bounded continuous functions on $\mathcal{X}$. However, it does not yield any useful finite sample estimate. On the other hand, if we take $\mathcal{F}$ to be a reproducing kernel Hilbert space (RKHS) associated with a symmetric positive semi-definite kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ such that $\int \sqrt{k(x, x)} dP(x) < \infty$ for all $P \in \mathcal{P}$, an equivalent closed form expression of the square of the MMD can be obtained as

$$\begin{aligned} d(P, Q) &= \int \int k(x, x') dP(x) dP(x') \\ &\quad + \int \int k(y, y') dQ(y) dQ(y') \\ &\quad - 2 \int \int k(x, y) dP(x) dQ(y). \end{aligned}$$

See, for example, [28]. Hereafter, the square of the MMD between two distributions P and Q will be denoted by $d(P, Q)$. This form easily allows an unbiased finite sample estimate. The choices of k for which the metric property holds are called characteristic kernels. For finite dimensional $\mathcal{X}$, several characteristic kernels have been well-studied (see [29]), while for infinite dimensional Hilbert spaces, Gaussian, Laplacian kernels have been shown to be characteristic (see [30]).

We can estimate MMD in the finite sample case as follows. Suppose $X = (x_1, x_2, \dots, x_{n_1})$ and $Y = (y_1, y_2, \dots, y_{n_2})$ are two ensembles of i.i.d. observations drawn from the distributions P and Q respectively. Then a V-statistic type estimate of $d(P, Q)$

is given by

$$\begin{aligned} d(\hat{P}, \hat{Q}) &= \frac{1}{n_1^2} \sum_{i, i'=1}^{n_1} k(x_i, x_{i'}) + \frac{1}{n_2^2} \sum_{j, j'=1}^{n_2} k(y_j, y_{j'}) \\ &\quad - \frac{2}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} k(x_i, y_j), \end{aligned}$$

where $\hat{P} = n_1^{-1} \sum_{i=1}^{n_1} \delta_{x_i}$ and $\hat{Q} = n_2^{-1} \sum_{j=1}^{n_2} \delta_{y_j}$ are the empirical measures. In the next section, we will develop novel clustering techniques using the MMD, which will allow us to identify clusters having different probability distributions.

III. CLUSTERING METHODOLOGY

Consider a sample $\mathcal{D} = \{X_1, X_2, \dots, X_n\}$ taking values in the ambient space $\mathcal{X}$. The goal is to partition $\mathcal{D}$ into K (may or may not be specified) groups such that the each group is as homogeneous as possible, and two distinct groups are as heterogeneous as possible. Mathematically, each group should have all observations from one single population/distribution and observations from any two distinct groups should belong to two distinct populations/distributions. To achieve this end, we need to have a measure of discrepancy between two groups/clusters. Consider two groups of observations $A = \{a_1, a_2, \dots, a_{n_1}\}$ and $B = \{b_1, b_2, \dots, b_{n_2}\}$. A measure of discrepancy between the sets A and B is given by

$$\begin{aligned} d_w(A, B) &= \frac{n_1 n_2}{n_1 + n_2} d(\hat{P}_A, \hat{P}_B) \\ &= \frac{n_1 n_2}{n_1 + n_2} \left\{ \frac{1}{n_1^2} \sum_{i, i'=1}^{n_1} k(a_i, a_{i'}) \right. \\ &\quad \left. + \frac{1}{n_2^2} \sum_{j, j'=1}^{n_2} k(b_j, b_{j'}) - \frac{2}{n_1 n_2} \sum_{i=1}^{n_1} \sum_{j=1}^{n_2} k(a_i, b_j) \right\}, \end{aligned}$$

where $\hat{P}_A$ and $\hat{P}_B$ are the empirical distributions of the observations of A and B respectively. Throughout this paper, we will take k to be a characteristic kernel. Since k is characteristic kernel, $d(\hat{P}_A, \hat{P}_B) \geq 0$ with equality holding if and only if $\hat{P}_A = \hat{P}_B$, i.e., $A = B$. Hence, $d_w(A, B) \geq 0$ and equality holds if and only if $A = B$. Here, the extra factor involving the cluster sizes acts as a balancing term to counter the situation where the group/cluster sizes may be particularly small (see [31]). If we have observations from two populations (i.e., $K = 2$), then the goal is to partition $\mathcal{D}$ into subsets A and B such that $d_w(A, B)$ is maximum. For a general K class problem, we would like to partition $\mathcal{D}$ into K subsets $A_1, A_2, \dots, A_K$ such that $d_w(A_i, A_j)$ is maximized for all $i \neq j$ and $i, j \in \{1, 2, \dots, K\}$.

We start with the case when $K = 2$. Since the objective is to get two homogeneous clusters, we start by selecting two subsets C_1 and C_2 of $\mathcal{D}$ where C_1 contains a single observation for which the d_w distance between C_1 and C_2 is maximum. This step automatically ensures that C_1 is homogeneous. The next step would be to add to C_1 those observations from C_2 which belong to the same population as the observation already in C_1 .

Algorithm BS: Binary Splitting.**Input:** Dataset $\mathcal{E}$.Set $C_1^{(0)} = \emptyset$ and $C_2^{(0)} = \mathcal{E}$.

- 1) for $r = 1, 2, \dots, |\mathcal{E}| - 1$ do
- 2) Find an element c from $C_2^{(r-1)}$ which maximizes $d_w(C_1^{(r-1)} \cup \{c\}, C_2^{(r-1)} \setminus \{c\})$.
- 3) $C_1^{(r)} = C_1^{(r-1)} \cup \{c\}$ and $C_2^{(r)} = C_2^{(r-1)} \setminus \{c\}$.
- 4) endfor

Output: $N_{max} := \arg \max_{r=1,2,\dots,|\mathcal{E}|-1} d_w(C_1^{(r)}, C_2^{(r)}), C_1^{(N_{max})}$
and $C_2^{(N_{max})}$

To achieve this, we start transferring observations from C_2 to C_1 one at a time such that the d_w -distance between C_1 and C_2 is maximum at each step. Formally, suppose that $C_i^{(r)}$ denotes the i -th set after the r -th iteration, $r = 1, 2, \dots, n - 1$ and $i = 1, 2$. After transferring observations by maximizing the d_w -distance at each iteration, we consider $C_1^{(N_{max})}$ and $C_2^{(N_{max})}$ as the final clusters where

$$N_{max} = \arg \max_{r=1,2,\dots,n-1} d_w(C_1^{(r)}, C_2^{(r)}). \quad (1)$$

This procedure, referred to as the Binary Splitting (BS) algorithm, is detailed above for any given dataset $\mathcal{E} \subseteq \mathcal{D}$.

The following Remark 1 states some interesting properties of the d_w -distance which facilitates computation of the Algorithm BS.

Remark 1: Suppose that in the $(r + 1)$ -th iteration of the Algorithm BS, the observation c is transferred from $C_2^{(r)}$ to $C_1^{(r)}$, so that $C_2^{(r+1)} = C_2^{(r)} \setminus \{c\}$ and $C_1^{(r+1)} = C_1^{(r)} \cup \{c\}$. Then, it can be shown that

$$\begin{aligned} d_w(C_1^{(r+1)}, C_2^{(r+1)}) - d_w(C_1^{(r)}, C_2^{(r)}) \\ = d_w(C_2^{(r)} \setminus \{c\}, \{c\}) - d_w(C_1^{(r)}, \{c\}). \end{aligned} \quad (2)$$

This identity has a two-fold implication:

- a) The change in the d_w -distance between the r -th and the $(r + 1)$ -th iteration only depends on the element c and the two sets in the r -th iteration. If c is closer to $C_1^{(r)}$ than $C_2^{(r)} \setminus \{c\}$, then the d_w -distance increases from the r -th to the $(r + 1)$ -th iteration and vice-versa.
- b) The computation of $d_w(C_1^{(r+1)}, C_2^{(r+1)})$ can be easily done by adding $d_w(C_1^{(r)}, C_2^{(r)})$ to the difference in the right hand side of (2). The latter computation is cheaper since one of the sets involved is a singleton.

When the number of populations K is larger than two, the Algorithm BS has must be applied repeatedly. The details will be given in the next subsection. We will propose clustering algorithms for both situation: (a) when the number of populations is unspecified, and (b) when it is specified. In the former case, our algorithm will automatically provide a bonafide estimator of the number of populations.

Algorithm SCC: Single Cluster Checking.**Input:** Dataset $\mathcal{E}$.Set $C_1^{(0)} = \emptyset$ and $C_2^{(0)} = \mathcal{E}$.

- 1) for $r = 1, 2, \dots, |\mathcal{E}| - 1$ do
- 2) Find an element c from $C_2^{(r-1)}$ which maximizes $d_w(C_1^{(r-1)} \cup \{c\}, C_2^{(r-1)} \setminus \{c\})$.
- 3) Set $C_1^{(r)} = C_1^{(r-1)} \cup \{c\}$ and $C_2^{(r)} = C_2^{(r-1)} \setminus \{c\}$.
- 4) endfor

Set $V = \max_{r=1,2,\dots,|\mathcal{E}|-1} d_w(C_1^{(r)}, C_2^{(r)}) / \min_{r=1,2,\dots,|\mathcal{E}|-1} d_w(C_1^{(r)}, C_2^{(r)})$.

Set $N_{max} := \arg \max_{r=1,2,\dots,|\mathcal{E}|-1} d_w(C_1^{(r)}, C_2^{(r)})$ and $N_{min} := \arg \min_{r=1,2,\dots,|\mathcal{E}|-1, r \neq N_{max}} d_w(C_1^{(r)}, C_2^{(r)})$.

Set $H = N_{max}(|\mathcal{E}| - N_{max}) / [N_{min}(|\mathcal{E}| - N_{min})]$.

If $N_{min} \leq N_{max}$ set $R = N_{max}(|\mathcal{E}| - N_{min}) / [N_{min}(|\mathcal{E}| - N_{max})]$

else set $R = N_{min}(|\mathcal{E}| - N_{max}) / [N_{max}(|\mathcal{E}| - N_{min})]$.

Output: If $|V - 1| < |V - (R/H)|$ out = 0 (Accept) else out = 1 (Reject).

A. Methodology for Unspecified Number of Populations

Since the number of populations is unspecified, we will have to first check whether the sample contains observations from more than one distribution or not. To achieve this goal, we propose an algorithm, referred to as the Algorithm SCC, to detect whether a dataset can be ascertained as a single cluster.

The above algorithm can be motivated as follows: if the sample comes from a single population, then one expects the sample MMD curve to be almost flat and V defined in Algorithm SCC to be close to 1. Consequently, if $|V - 1|$ exceeds a certain threshold, one would conclude the presence of more than one populations in the data. The novelty of Algorithm SCC is that we use a data-driven threshold rather than an arbitrary prefixed value. The rationale behind choosing the threshold as $|V - (R/H)|$ is discussed through an oracle analysis in Section IV.

If the Algorithm SCC determines that the entire dataset consists of observations from more than one populations, we will have to apply the Algorithm BS and the Algorithm SCC repeatedly to split any subset of the whole data into two disjoint subsets and also check whether each of these two subsets conform to the single population hypothesis. Now, we describe the algorithm, referred to as Clustering Using Recursive Binary Splitting (CURBS)-I, for clustering the entire dataset. We will discuss the justification of this algorithm through an oracle analysis in Section IV.

B. Methodology for Specified Number of Populations

In case the number of populations J is specified in advance (may or may not be the true number of populations K), we will have to modify the Algorithm CURBS-I to accommodate this information. More specifically, in place of the Algorithm SCC which was applied to all the clusters obtained in a given stage (cf. steps 3 and 4 of the Algorithm CURBS-I), we deliberately split each of these clusters using the Algorithm BS. Consequently, it may so happen that some of the clusters containing observations from a single population is split into two sub-clusters which now

Algorithm: CURBS-I.**Input:** Dataset $\mathcal{D}$.

- 1) Apply Algorithm SCC to $\mathcal{D}$.
- 2) If Accepted, END, else apply Algorithm BS to $\mathcal{D}$.
- 3) Apply Algorithm SCC to the output clusters obtained.
If Algorithm SCC is Accepted for a cluster, keep that cluster unchanged, else apply Algorithm BS on that cluster and obtain two new clusters.
- 4) Repeat Step 3 until Algorithm SCC is Accepted for each of the clusters obtained.

Output: $\hat{K}$ and the clusters $C_1, C_2, \dots, C_{\hat{K}}$, where $\hat{K}$ is the number of clusters finally obtained.**Algorithm: CURBS-II.****Input:** Dataset $\mathcal{D}$ and specified number $J (> 1)$ of clusters.

- 1) Apply Algorithm BS to $\mathcal{D}$.
- 2) If $J = 2$ then END, else set $i = 2$.
- 3) Apply Algorithm BS to each of the i clusters obtained.
- 4) Compute all possible pairwise d_w -distances between the $2i$ clusters obtained in Step 3 and arrange them in the increasing order of magnitude.
- 5) Merge those $(i - 1)$ pair(s) of clusters obtained in Step 3 having the least d_w -distances.
- 6) If $i = J - 1$ END, else update $i = i + 1$ and go to Step 3.

Output: The clusters $C_1, C_2, \dots, C_J$.

needs to be merged. This merging is done based on d_w -distances between sub-clusters, and it also allows us to control the number of output clusters using the information about J . The details of such an algorithm, referred to as the Algorithm CURBS-II, is given above. We will discuss the justification of this algorithm through an oracle analysis in Section IV.

Remark 2: Algorithm BS and Algorithm SCC both require computing the sample MMD curve (Steps 1–4), which can be done in $O(n^2)$ time for a sample of size n . Hence, both have $O(n^2)$ time complexity. Algorithm CURBS-I repeatedly applies Algorithm SCC (and Algorithm BS when Algorithm SCC rejects). In all numerical studies in the paper, the number of times Algorithm SCC is called depends only on the true number of clusters K , yielding an overall complexity of $O(n^2)$. In the unlikely worst case of $O(n)$ repeated rejections by Algorithm SCC, the complexity can increase to $O(n^3)$. Algorithm CURBS-II requires Algorithm BS only a finite number of times determined by the prescribed number of clusters J , and therefore also has $O(n^2)$ time complexity.

IV. ORACLE ANALYSIS OF PROPOSED CLUSTERING ALGORITHMS

This section is aimed at providing a motivation and justification for the algorithms detailed in Section 3. The analysis done here will be based on an oracle situation which will be described shortly. To this end, consider first the case when the observations come from two different distributions. Denote the observations

by $X_1, X_2, \dots, X_{n_1}$ from a distribution P_1 and $Y_1, Y_2, \dots, Y_{n_2}$ from a distribution P_2 . Let us denote the empirical distributions of the X_i 's and the Y_j 's by $\hat{P}_1$ and $\hat{P}_2$, respectively. Suppose we have two clusters $S_1 = \{X_1, \dots, X_{m_1}, Y_1, \dots, Y_{m_2}\}$ and $S_2 = \{X_{m_1+1}, \dots, X_{m_1+m_3}, Y_{m_2+1}, \dots, Y_{m_2+m_4}\}$, for $m_1 + m_3 \leq n_1$ and $m_2 + m_4 \leq n_2$. As opposed to the usual empirical distributions of the sets S_1 and S_2 , we shall define the “representative distributions” of the sets S_1 and S_2 . Note that S_1 can be considered as a possible realization from $\alpha\hat{P}_1 + (1 - \alpha)\hat{P}_2$ for some $\alpha \in [0, 1]$. The most canonical choice of α is its natural estimator, namely, the sample proportion of observations in S_1 coming from P_1 , i.e., $m_1/(m_1 + m_2)$. The representative distribution of S_1 is defined as $\tilde{P}_{S_1} = m_1/(m_1 + m_2)\hat{P}_1 + m_2/(m_1 + m_2)\hat{P}_2$. Note in passing that if we draw a random sample of size $(m_1 + m_2)$ from $\tilde{P}_{S_1}$, the event that the outcome will be S_1 has a positive probability. The representative distribution of S_2 , denoted by $\tilde{P}_{S_2}$, is given by $\tilde{P}_{S_2} = m_3/(m_3 + m_4)\hat{P}_1 + m_4/(m_3 + m_4)\hat{P}_2$. Observe that even if S_1 has some other configuration of m_1 elements from $\hat{P}_1$ and m_2 elements from $\hat{P}_2$, its representative distribution remains unchanged while the empirical distribution changes. This property of representative distributions allows us to capture the intrinsic composition of a set of observations. Note that in a clustering problem, the degree of homogeneity/heterogeneity of a cluster is determined not by which elements of each population are present in the cluster, but by the proportions of elements corresponding to the different populations (that is, the intrinsic composition of a cluster). This motivates us to use the representative distributions in the oracle analysis. In case the observations come from $K (> 2)$ different distributions, the definitions of the representative distributions can be modified similarly.

Our oracle analysis will assume the knowledge of the empirical distributions, thus converting the unsupervised problem into a supervised problem. This will imply that for any cluster, we will be able to construct its representative distribution. The theoretical analyses of this section shows that the full understanding of the near-perfect clustering achieved by the Algorithms CURBS-I and CURBS-II can be effectively obtained through the perspective of representative distributions. More specifically, we provide oracle analogs of the algorithms in Section III using a version of the MMD distance between representative distributions. These oracle analogs of the Algorithms CURBS-I and CURBS-II will be theoretically shown to have perfect clustering property. Indeed, the algorithms in Section III should be viewed as data-driven versions of these oracle algorithms.

Remark 3: One of the key components in the proof of the perfect clustering behaviour of the oracle algorithms is the following remarkable property of MMD. If F and G are two probability distributions and $\alpha, \beta \in [0, 1]$, then

$$d(\alpha F + (1 - \alpha)G, \beta F + (1 - \beta)G) = (\alpha - \beta)^2 d(F, G). \quad (3)$$

The above equation implies that the square of the MMD between a pair of mixture distributions with two components will always

Algorithm BS*: Oracle Binary Splitting.**Input:** Dataset $\mathcal{E}$.Set $C_1^{(0)} = \emptyset$ and $C_2^{(0)} = \mathcal{E}$.

- 1) for $r = 1, 2, \dots, |\mathcal{E}| - 1$, do
- 2) Find an element c from $C_2^{(r-1)}$ which maximizes $d_w^*(C_1^{(r-1)} \cup \{c\}, C_2^{(r-1)} \setminus \{c\})$.
- 3) Set $C_1^{(r)} = C_1^{(r-1)} \cup \{c\}$ and $C_2^{(r)} = C_2^{(r-1)} \setminus \{c\}$.
- 4) endfor

Output: $N_{max}^* := \arg \max_{r=1,2,\dots,|\mathcal{E}|-1} d_w^*(C_1^{(r)}, C_2^{(r)})$, $C_1^{(N_{max}^*)}$
and $C_2^{(N_{max}^*)}$.

be less than or equal to the square of the MMD between the two component distributions with equality holding if and only if $(\alpha, \beta) = (0, 1)$ or $(1, 0)$. Indeed, we are not aware of any other probability metric which enjoys this property, and this motivated us to use MMD in our clustering algorithms. Since the representative distributions are also mixture distributions, the clustering algorithms are designed by carefully utilizing the property in (3). Moreover, all the theoretical results of this paper also heavily rely on the same property.

Now, we formally describe the oracle algorithms starting with a measure of dissimilarity between two clusters. An oracle version of the d_w^* -distance between two clusters S_1 and S_2 is given by

$$d_w^*(S_1, S_2) = \frac{|S_1| |S_2|}{|S_1| + |S_2|} d(\tilde{P}_{S_1}, \tilde{P}_{S_2}),$$

where for any set S , its cardinality is denoted by $|S|$. Note that the d_w^* measure defined above uses the representative distributions of the two sets whereas the d_w -distance uses the empirical distributions of the two sets. The oracle analog of the Algorithm BS, referred to as the Algorithm BS*, is detailed above.

Observe that in Step 2, the choice of the observation c is not unique. Indeed, if an observation from P_1 is a maximizer, then so is any other observation from P_1 in the set C_2 . We will now discuss the oracle versions of the other proposed algorithms.

A. Oracle Version of Algorithm CURBS-I

In the Algorithm CURBS-I, the question of whether a cluster has observations from a single population was addressed using the Algorithm SCC. In the oracle scenario, single cluster checking can be done simply by investigating the values of d_w^* over different iterations as in the Algorithm BS*. If a cluster contains observations from a single distribution, then the values of d_w^* over all iterations will be zero. Otherwise, the maximum value of d_w^* will be greater than its minimum value. Recall that in the Algorithm SCC, we accepted that the cluster has observations from more than one distribution if V is closer to R/H than 1. This can now be motivated as follows. Let us define the oracle versions of R and H as given above when the dataset $\mathcal{D}$ consists

of n observations:

$$R^* = \begin{cases} \frac{N_{max}^*(n - N_{min}^*)}{N_{min}^*(n - N_{max}^*)} & \text{if } N_{min}^* < N_{max}^* \\ \frac{N_{min}^*(n - N_{max}^*)}{N_{max}^*(n - N_{min}^*)} & \text{if } N_{min}^* > N_{max}^* \end{cases} \quad (4)$$

$$H^* = \frac{N_{max}^*(n - N_{max}^*)}{N_{min}^*(n - N_{min}^*)},$$

where $C_1^{(r)}$'s and $C_2^{(r)}$'s are as in the Algorithm BS*, and

$$N_{max}^* := \arg \max_{1 \leq r \leq n-1} d_w^*(C_1^{(r)}, C_2^{(r)}),$$

$$N_{min}^* := \arg \min_{1 \leq r \leq n-1} d_w^*(C_1^{(r)}, C_2^{(r)}). \quad (5)$$

In case the number of populations $K > 1$, the oracle version of V is defined as

$$V^* = \frac{\max_{1 \leq r \leq n-1} d_w^*(C_1^{(r)}, C_2^{(r)})}{\min_{1 \leq r \leq n-1} d_w^*(C_1^{(r)}, C_2^{(r)})}. \quad (6)$$

Theorem 1: Consider a set of n observations of sizes $n_1, n_2, \dots, n_K$ from $K > 1$ populations. The following properties are true:

- a) $V^* = R^*$,
- b) there exists $n_0 \geq 1$ such that $N_{min}^* = n - 1$ for all $n \geq n_0$ provided that $\liminf_{n \rightarrow \infty} (n_i/n) > 0$, for all $i = 1, 2, \dots, K$, and
- c) under the condition in part (b), there exists $n^* \geq n_0$ such that $H^* > 1$ for $n \geq n^*$.

The quantities n_0 and n^* in the previous theorem depend on the inter-population MMD distances between the K populations and also the proportions of each population present in the sample (see the proof of Theorem 1 in the Supplementary Material). Since $R^* > H^*$ (which follows from (4) and (5)), it can be deduced from Theorem 1 that when the number of populations $K > 1$, we have

$$|V^* - (R^*/H^*)| = |R^* - (R^*/H^*)|$$

$$= R^* - (R^*/H^*) < R^* - 1 = |V^* - 1|. \quad (7)$$

If the cluster has observations from a single distribution, then the numerator and the denominator of V^* are both equal to zero. In the empirical situation, it is observed that the numerator and the denominator of V^* are close to each other, and hence, we use the convention that $V^* = 1$ when the observations come from a single distribution. Thus, since $R^* > H^*$, we have $|V^* - (R^*/H^*)| > |V^* - 1|$. These discussions inspire an oracle algorithm, referred to as the Algorithm SCC*. The empirical analog of this oracle algorithm is the Algorithm SCC given in Section III.

Observe that in view of Theorem 1 and the discussion after (7), the decision made by the Algorithm SCC* is always correct for all sufficiently large sample sizes. We now provide the oracle version of the Algorithm CURBS-I, referred to as the Algorithm CURBS-I*, by applying the Algorithms BS* and SCC* repeatedly.

Algorithm SCC*: Oracle Single Cluster Checking.**Input:** Dataset $\mathcal{E}$.Set $C_1^{(0)} = \emptyset$ and $C_2^{(0)} = \mathcal{E}$.1) for $r = 1, 2, \dots, |\mathcal{E}| - 1$, do2) Find an element c from $C_2^{(r-1)}$ which maximizes

$$d_w^*(C_1^{(r-1)} \cup \{c\}, C_2^{(r-1)} \setminus \{c\}).$$

3) Set $C_1^{(r)} = C_1^{(r-1)} \cup \{c\}$ and $C_2^{(r)} = C_2^{(r-1)} \setminus \{c\}$.

4) endfor

Set .

Set $N_{max}^* := \arg \max_{1 \leq r \leq |\mathcal{E}| - 1} d_w^*(C_1^{(r)}, C_2^{(r)})$ and $N_{min}^* :=$

$$\arg \min_{1 \leq r \leq |\mathcal{E}| - 1, r \neq N_{max}^*} d_w^*(C_1^{(r)}, C_2^{(r)}).$$

Set $H^* = N_{max}^* (|\mathcal{E}| - N_{max}^*) / [N_{min}^* (|\mathcal{E}| - N_{min}^*)]$.If $N_{min}^* < N_{max}^*$, set

$$R^* = N_{max}^* (|\mathcal{E}| - N_{min}^*) / [N_{min}^* (|\mathcal{E}| - N_{max}^*)] \text{ else set}$$

$$R^* = N_{min}^* (|\mathcal{E}| - N_{max}^*) / [N_{max}^* (|\mathcal{E}| - N_{min}^*)].$$

Output: If $|V^* - 1| < |V^* - (R^*/H^*)|$ out = 0 (Accept) else out = 1 (Reject).**Algorithm:** CURBS-I*.**Input:** Dataset $\mathcal{D}$.1) Apply Algorithm SCC* to $\mathcal{D}$.2) If Accepted, END, else apply Algorithm BS* to $\mathcal{D}$.

3) Apply Algorithm SCC* to the output clusters obtained. If Algorithm SCC* is Accepted for a cluster, keep that cluster unchanged, else apply Algorithm BS* on that cluster and obtain two new clusters.

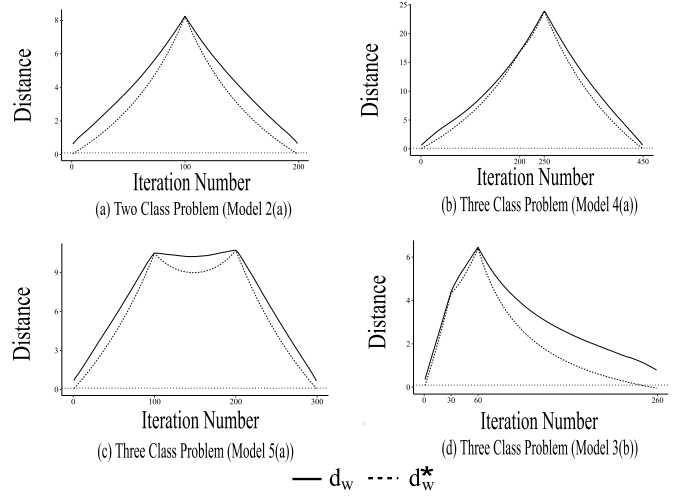
4) Repeat Step 3 until Algorithm SCC* is Accepted for each of the clusters obtained.

Output: K and the clusters $C_1, C_2, \dots, C_K$, where K is the number of clusters finally obtained.

Next, we discuss the behaviour of the Algorithm CURBS-I* for different values of the number of populations K . The backbone of this analysis is the understanding of the values of d_w^* over different iterations as in the Algorithm BS*.

Case 1 ($K = 2$): Denote the two empirical distributions by $\hat{P}_1$ and $\hat{P}_2$, which have been constructed based on samples of sizes n_1 and n_2 , respectively. Denote $n = n_1 + n_2$. As in the Algorithm BS*, if we start with two sets C_1 and C_2 such that C_1 is empty and C_2 contains all the observations, then an observation from $\hat{P}_1$ will be transferred to C_1 in the first iteration if and only if $n_1 \leq n_2$ (See part (a) of Lemma 1 in the Supplementary Material). Without loss of generality, let us assume that an observation from $\hat{P}_1$ is transferred to C_1 in the first iteration, and as such $n_1 \leq n_2$. Then, till the n_1 -th iteration, observations only from $\hat{P}_1$ will be transferred to C_1 at every iteration and the d_w^* curve will keep on increasing (See part (a) of Lemma 1 in the Supplementary Material). In fact, it can be shown that

$$\begin{aligned} d_w^*(C_1^{(r)}, C_2^{(r)}) &= \begin{cases} \frac{rn_2^2}{n(n-r)} d(\hat{P}_1, \hat{P}_2); & \text{if } r = 1, 2, \dots, n_1, \\ \frac{n_1^2(n-r)}{rn} d(\hat{P}_1, \hat{P}_2); & \text{if } r = n_1 + 1, \dots, n. \end{cases} \end{aligned} \quad (8)$$

Fig. 1. Curves of d_w and d_w^* .

We have plotted the d_w^* curve (given by (8)) in plot (a) of Fig. 1 (dashed curve) with the value of $d(\hat{P}_1, \hat{P}_2)$ as given by Model (2a) in Section V for sample sizes $n_1 = n_2 = 100$. We have also provided the d_w curve for the above model (solid curve) in the same plot, and it is seen that the two curves are quite similar in shape. This similarity plays a crucial role in the near-perfect clustering performance of the Algorithm CURBS-I observed for the above model (see Table I).

Case 2 ($K = 3$): Denote the three empirical distributions by $\hat{P}_1, \hat{P}_2$ and $\hat{P}_3$, which have been constructed based on samples of sizes n_1, n_2 and n_3 , respectively. Denote $n = n_1 + n_2 + n_3$. We start with two sets C_1 and C_2 such that C_1 is empty and C_2 contains all observations and continue transferring the observations from C_2 to C_1 . An observation from $\hat{P}_1$ will be transferred to C_1 in the first iteration if and only if the following two conditions hold (see part (a) of Lemma 2 in the Supplementary Material).

$$(A1) \quad (n_2 - n_1)d(\hat{P}_1, \hat{P}_2) + n_3 \{d(\hat{P}_1, \hat{P}_3) - d(\hat{P}_2, \hat{P}_3)\} \geq 0$$

$$(A2) \quad (n_3 - n_1)d(\hat{P}_1, \hat{P}_3) + n_2 \{d(\hat{P}_1, \hat{P}_2) - d(\hat{P}_2, \hat{P}_3)\} \geq 0$$

Under the same two conditions (A1) and (A2), it can be ensured that till the n_1 -th iteration, observations only from $\hat{P}_1$ will be transferred to C_1 at each iteration and the d_w^* curve will keep on increasing (see part (a) of Lemma 2 in the Supplementary Material). In the balanced case (when $n_1 = n_2 = n_3$), the conditions (A1) and (A2) boils down to requiring

$$d(\hat{P}_2, \hat{P}_3) \leq \min\{d(\hat{P}_1, \hat{P}_2), d(\hat{P}_1, \hat{P}_3)\},$$

i.e., the distributions $\hat{P}_2$ and $\hat{P}_3$ are closer to each other than to $\hat{P}_1$. This also explains why at the first step $\hat{P}_1$ gets separated from the cluster containing observations from $\hat{P}_2$ and $\hat{P}_3$. Next, assuming conditions (A1) and (A2), an observation from $\hat{P}_2$ will be transferred to C_1 after the n_1 -th iteration if and only if the following condition holds (see part (b) of Lemma 2 in the Supplementary Material).

$$(A3) \quad n_1(n_2 + n_3 - 1)\{d(\hat{P}_1, \hat{P}_3) - d(\hat{P}_1, \hat{P}_2)\} + (n_3 - n_2)(n_1 + 1)d(\hat{P}_2, \hat{P}_3) \geq 0.$$

It can also be shown that under the assumptions (A1), (A2) and (A3), all the observations from $\hat{P}_2$ will be transferred to C_1 in the subsequent iterations till the $(n_1 + n_2)$ -th iteration (see part (b) of Lemma 2 in the Supplementary Material). In the balanced case, the above condition reduces to requiring $d(\hat{P}_1, \hat{P}_3) \geq d(\hat{P}_1, \hat{P}_2)$, i.e., $\hat{P}_2$ is closer to $\hat{P}_1$ than $\hat{P}_3$. This condition is expected since $\hat{P}_2$ being closer to $\hat{P}_1$ than $\hat{P}_3$ would naturally imply that it should be the group that first gets merged to $\hat{P}_1$. Further, one can explicitly derive the form of the d_w^* curve over all iterations (see Lemma 2 in the Supplementary Material), namely,

$$d_w^*(C_1^{(r)}, C_2^{(r)}) = \begin{cases} \frac{r}{n(n-r)} \{n_2(n_2 + n_3)d(\hat{P}_1, \hat{P}_2) + n_3(n_2 + n_3)d(\hat{P}_1, \hat{P}_3) - n_2n_3d(\hat{P}_2, \hat{P}_3)\}; & \text{if } r = 1, 2, \dots, n_1, \\ \frac{nn_1 - r(n_1 + n_3)}{n} \left\{ \frac{n_1}{r} d(\hat{P}_1, \hat{P}_2) - \frac{n_3}{n-r} d(\hat{P}_2, \hat{P}_3) \right\} + \frac{n_1n_3}{n} d(\hat{P}_1, \hat{P}_3); & \text{if } r = n_1 + 1, \dots, n_1 + n_2, \\ \frac{n-r}{rn} \{n_2(n_1 + n_2)d(\hat{P}_2, \hat{P}_3) + n_1(n_1 + n_2)d(\hat{P}_1, \hat{P}_3) - n_1n_2d(\hat{P}_1, \hat{P}_2)\}; & \text{if } r = n_1 + n_2 + 1, \dots, n. \end{cases} \quad (9)$$

As mentioned earlier, the above formula reveals that the d_w^* curve keeps on increasing till the n_1 -th iteration. Moreover, the d_w^* curve keeps on decreasing after the $(n_1 + n_2)$ -th iteration. Unlike the case $K = 2$, the d_w^* curve may increase or decrease between the n_1 -th and the $(n_1 + n_2)$ -th iterations depending on the sample sizes and the values of the pairwise MMD between the three empirical distributions. It can be shown that between the n_1 -th and the $(n_1 + n_2)$ -th iterations, the d_w^* curve will be a convex function of the iteration number r (See part (b) of Lemma 2 in the Supplementary Material). From (9), doing some simple algebraic calculations, it can be shown that for $r = n_1 + 1, n_1 + 2, \dots, n_1 + n_2$,

$$d_w^*(C_1^{(r)}, C_2^{(r)}) - d_w^*(C_1^{(r-1)}, C_2^{(r-1)}) = \frac{n_3^2}{(n-r)(n-r+1)} d(\hat{P}_2, \hat{P}_3) - \frac{n_1^2}{r(r-1)} d(\hat{P}_1, \hat{P}_2), \quad (10)$$

which is an increasing function of r . Thus, one possible scenario is when it is increasing, and this holds if and only if $d_w^*(C_1^{(n_1+1)}, C_2^{(n_1+1)}) > d_w^*(C_1^{(n_1)}, C_2^{(n_1)})$ (as the right hand side of (10) is an increasing function of iteration number). One can also write the previous inequality in the following equivalent form:

$$\frac{n_1}{n_1 + 1} d(\hat{P}_1, \hat{P}_2) < \frac{n_3^2}{(n_2 + n_3)(n_2 + n_3 - 1)} d(\hat{P}_2, \hat{P}_3). \quad (11)$$

Observe that in the balanced case, the above relation (11) cannot hold in view of the Assumptions (A1), (A2) and (A3). Thus, the d_w^* curve cannot increase after the n_1 -th iteration in the balanced case. Another possible scenario is when the d_w^* curve is decreasing after the n_1 -th iteration which holds if and only if $d_w^*(C_1^{(n_1+n_2)}, C_2^{(n_1+n_2)}) < d_w^*(C_1^{(n_1+n_2-1)}, C_2^{(n_1+n_2-1)})$ (as the right hand side of (10) is an increasing function of iteration number). The following condition is equivalent to the previous

inequality:

$$\frac{n_3}{n_3 + 1} d(\hat{P}_2, \hat{P}_3) < \frac{n_1^2}{(n_1 + n_2)(n_1 + n_2 - 1)} d(\hat{P}_1, \hat{P}_2). \quad (12)$$

In the balanced case, the above relation (12) holds if and only if $d(\hat{P}_1, \hat{P}_2) > 4d(\hat{P}_2, \hat{P}_3)$, using $n_3 + 1 \approx n_3$ and $n_1 + n_2 + 1 \approx n_1 + n_2$. In general, if neither of the conditions (11) and (12) holds, the d_w^* curve will be a U-shaped curve between the n_1 -th and the $(n_1 + n_2)$ -th iterations.

We have plotted the d_w^* curve (given by (9)) with the values of the square MMDs between the three empirical distributions as given by each of the three models, namely, Model (5a), Model (7a) and Model (5b) given in Section V (see the plots (b), (c) and (d) of Fig. 1). The sample sizes for Model (5a) and Model (7a) are taken to be $n_1 = n_2 = n_3 = 100$, while for Model (5b), we have taken $n_1 = 30, n_2 = 30, n_3 = 200$. In Model (5a), neither of the conditions (11) and (12) holds, and therefore, d_w^* will be U-shaped between the n_1 -th and the $(n_1 + n_2)$ -th iterations. The populations chosen for Model (7a) satisfies condition (12), and hence the d_w^* curve will decrease between the n_1 -th and the $(n_1 + n_2)$ -th iterations. The unequal sample sizes chosen for Model (5b) ensures that the condition (11) is satisfied, in which case, the d_w^* curve will be increasing between the n_1 -th and the $(n_1 + n_2)$ -th iterations. We have also plotted the d_w curves corresponding to each of the above models with their respective d_w^* curves. The marked similarity in shapes of plays a crucial role in the near-perfect clustering performance of the Algorithm CURBS-I (see Table I).

Case 3 ($K > 3$): Denote the K empirical distributions by $\hat{P}_1, \dots, \hat{P}_K$, which have been constructed based on samples of sizes $n_1, \dots, n_K$, respectively. We start with two sets C_1 and C_2 such that C_1 is empty and C_2 contains all observations and continue transferring the observations from C_2 to C_1 . Consider the iteration where C_1 contains observations from $\hat{P}_1, \dots, \hat{P}_m$ for some $1 \leq m \leq K - 2$ (without loss of generality). A crucial observation is that the analysis of the next iteration can be viewed from the perspective of a three-population problem considered earlier: the mixtures of the distributions in C_1 plays the role of the first population, the distribution of the candidate to be transferred from C_2 to C_1 plays the role of the second population, and the mixture of the remaining distributions in C_2 plays the role of the third population. This analogy with the three-population problem is also exploited to show that, at any step, the Algorithm BS* will produce two clusters such that for each $\hat{P}_j$, observations from this distribution will be contained in exactly one of the two clusters (see Lemma 3 in the Supplementary Material). Consequently, any cluster containing observations from a single distribution will no longer be split, while clusters containing observations from more than one population will continue to be split so long as each of the final clusters contain observations from a single distribution.

An observation from $\hat{P}_1$ will be transferred to C_1 in the first iteration if and only if the following condition holds

$$(A4) \quad (n^2 - n - nn_1 + 1) \sum_{i=1}^K n_i d(\hat{P}_1, \hat{P}_i) > (n^2 - n - nn_j + 1) \sum_{i=1}^K n_i d(\hat{P}_j, \hat{P}_i) \text{ for all } j = 2, 3, \dots, K.$$

Under the condition (A4), it can be ensured that till the n_1 -th iteration, observations only from $\hat{P}_1$ will be transferred to C_1 at each iteration, and the d_w^* curve will keep on increasing. In the balanced case (when $n_1 = n_2 = \dots = n_K$), the condition (A4) reduces to requiring $\sum_{i=1}^K d(\hat{P}_1, \hat{P}_i) > \sum_{i=1}^K d(\hat{P}_l, \hat{P}_i)$ for all $l = 2, 3, \dots, K$. For $l = 1, 2, \dots, K - 2$, after transferring all observations of $\hat{P}_1, \hat{P}_2, \dots, \hat{P}_l$ to C_1 (without any loss of generality) till iteration $n_1 + n_2 + \dots + n_l$, an observation from $\hat{P}_{l+1}$ will be transferred to C_1 in the next iteration if and only if the following condition holds:

$$(A5) \quad n \left(\sum_{i=l+1}^K n_i - 1 \right) \sum_{i=1}^l n_i \left\{ d(\hat{P}_i, \hat{P}_j) - d(\hat{P}_i, \hat{P}_{l+1}) \right\} \\ + \left(\sum_{i=1}^l n_i + 1 \right)^2 \sum_{i=l+1}^K n_i \left\{ (n_j - 1) d(\hat{P}_i, \hat{P}_j) \right. \\ \left. - (n_{l+1} - 1) d(\hat{P}_i, \hat{P}_{l+1}) \right\} + \left(\sum_{i=1}^l n_i + 1 \right) \\ \left(\sum_{i=l+1}^K n_i - 1 \right) \sum_{i=l+1}^K n_i \left\{ d(\hat{P}_i, \hat{P}_{l+1}) - d(\hat{P}_i, \hat{P}_j) \right\} > 0$$

for all $j = l + 2, l + 3, \dots, K$. Under the same assumptions, it can also be shown that observations only from $\hat{P}_{l+1}$ will be transferred to C_1 at every iteration until the $(n_1 + n_2 + \dots + n_{l+1})$ -th iteration. In the balanced case, using $\sum_{i=l+1}^K n_i - 1 \approx \sum_{i=l+1}^K n_i$ and $\sum_{i=1}^l n_i + 1 \approx \sum_{i=1}^l n_i$ in (A5), the condition (A5) reduces to the following condition: $(nl + K)(nl + K - K^2) \sum_{i=l+1}^K \{d(\hat{P}_i, \hat{P}_{l+1}) - d(\hat{P}_i, \hat{P}_j)\} + K^2 \{n(K - l) - K\} \sum_{i=1}^l \{d(\hat{P}_i, \hat{P}_j) - d(\hat{P}_i, \hat{P}_{l+1})\} > 0$ for all $j = l + 2, l + 3, \dots, K$.

The above condition is natural since an observation from $\hat{P}_{l+1}$ will be transferred to C_1 in the $(n_1 + n_2 + \dots + n_{l+1})$ -th iteration provided that it is closer to the mixture of $\hat{P}_1, \hat{P}_2, \dots, \hat{P}_l$ than the mixture of $\hat{P}_{l+2}, \dots, \hat{P}_K$.

One can explicitly derive the form of the d_w^* curve over all iterations, namely,

$$d_w^* \left(C_1^{(r)}, C_2^{(r)} \right) = \begin{cases} \frac{r(n-n_1)^2}{n(n-r)} \left\{ \frac{1}{n-n_1} \sum_{i=2}^K n_i d(\hat{P}_i, \hat{P}_1) \right. \\ \left. - \frac{1}{2(n-n_1)^2} \sum_{i,j=2}^K n_i n_j d(\hat{P}_i, \hat{P}_j) \right\}; & \text{if } r = 1, 2, \dots, n_1, \\ \frac{n(\sum_{i=1}^l n_i) - r(n-n_{l+1})}{n} \left\{ \frac{\sum_{i=1}^l n_i}{r} d \left(\frac{\sum_{i=1}^l n_i \hat{P}_i}{\sum_{i=1}^l n_i}, \hat{P}_{l+1} \right) \right. \\ \left. - \frac{\sum_{i=l+2}^K n_i}{n-r} d \left(\hat{P}_{l+1}, \frac{\sum_{i=l+2}^K n_i \hat{P}_i}{\sum_{i=l+2}^K n_i} \right) \right\} \\ + \frac{(\sum_{i=1}^l n_i)(\sum_{i=l+2}^K n_i)}{n} d \left(\frac{\sum_{i=1}^l n_i \hat{P}_i}{\sum_{i=1}^l n_i}, \frac{\sum_{i=l+2}^K n_i \hat{P}_i}{\sum_{i=l+2}^K n_i} \right); \\ \text{if } r = \sum_{i=1}^l n_i + 1, \dots, \sum_{i=1}^{l+1} n_i \text{ for } l = 1, 2, \dots, K - 2, \\ \frac{(n-r)}{rn} \{ (n - n_K) \sum_{i=1}^{K-1} n_i d(\hat{P}_i, \hat{P}_K) \\ - \frac{1}{2} \sum_{i,j=1}^{K-1} n_i n_j d(\hat{P}_i, \hat{P}_j) \}; & \text{if } r = \sum_{i=1}^{K-1} n_i + 1, \dots, n. \end{cases}$$

As mentioned earlier, the above formula reveals that the d_w^* curve keeps on increasing till the n_1 -th iteration and keeps on decreasing after the $(\sum_{i=1}^{K-1} n_i)$ -th iteration. Further, it can be shown that for each $l = 1, 2, \dots, K - 2$, the d_w^* curve will be a convex function of iteration number r between the $(\sum_{i=1}^{l-1} n_i)$ -th and the $(\sum_{i=1}^l n_i)$ -th iterations.

For $l = 1, 2, \dots, K - 2$, one possible scenario is that the d_w^* curve is increasing between the $(\sum_{i=1}^l n_i)$ -th and the $(\sum_{i=1}^{l+1} n_i)$ -th iterations. This holds if and only if the following condition is true:

$$\frac{\sum_{i=1}^l n_i}{\sum_{i=1}^l n_i + 1} d \left(\frac{\sum_{i=1}^l n_i \hat{P}_i}{\sum_{i=1}^l n_i}, \hat{P}_{l+1} \right) \\ < \frac{(\sum_{i=l+2}^K n_j)^2}{(\sum_{i=l+1}^K n_j)(\sum_{i=l+1}^K n_j - 1)} d \left(\hat{P}_{l+1}, \frac{\sum_{j=l+2}^K n_j \hat{P}_j}{\sum_{i=l+2}^K n_j} \right). \quad (13)$$

Another possible scenario is when d_w^* curve is decreasing between the $(\sum_{i=1}^l n_i)$ -th and the $(\sum_{i=1}^{l+1} n_i)$ -th iterations, and this holds if and only if the following condition is true:

$$\frac{\sum_{i=l+2}^K n_j}{\sum_{i=l+2}^K n_j + 1} d \left(\hat{P}_{l+1}, \frac{\sum_{j=l+2}^K n_j \hat{P}_j}{\sum_{i=l+2}^K n_j} \right) \\ < \frac{(\sum_{i=1}^l n_i)^2}{(\sum_{i=1}^{l+1} n_i)(\sum_{i=1}^{l+1} n_i - 1)} d \left(\frac{\sum_{i=1}^l n_i \hat{P}_i}{\sum_{i=1}^l n_i}, \hat{P}_{l+1} \right). \quad (14)$$

If neither (13) nor (14) hold, the d_w^* curve will be a U-shaped curve between the $(\sum_{i=1}^l n_i)$ -th and the $(\sum_{i=1}^{l+1} n_i)$ -th iterations. Hence, the d_w^* curve attains maximum at iteration $r = \sum_{i=1}^l n_i$ for some $l \in \{1, 2, \dots, K - 1\}$.

Remark 4: Now, we discuss the possible values of N_{max}^* for difference choices of K .

Case 1 ($K = 2$): It follows from the discussion of the behaviour of Algorithm BS* for $K = 2$ that $N_{max}^* = \min\{n_1, n_2\}$.

Case 2 ($K = 3$): If we assume that an observation from $\hat{P}_1$ is transferred to C_1 in the first iteration, and an observation from $\hat{P}_2$ is transferred to C_1 after the n_1 -th iteration, then it follows from (9) that $N_{max}^* \geq n_1$ and $N_{max}^* \leq n_1 + n_2$.

Case 3 ($K > 3$): If we assume that an observation from $\hat{P}_1$ is transferred to C_1 in the first iteration, and an observation from $\hat{P}_2$ is transferred to C_1 after the n_1 -th iteration, then it follows from (13) that $N_{max}^* \geq n_1$ and $N_{max}^* \leq \sum_{i=1}^{K-1} n_i$.

Now, let us define the perfectness of a clustering algorithm and then make the formal statement about the perfect clustering property of the Algorithm CURBS-I*.

Definition IV.1: For any fixed J , let $C_1, C_2, \dots, C_J$ be J clusters obtained using a clustering algorithm on $\mathcal{D} = \cup_{j=1}^K \mathcal{D}_j$, which consists of observations from K populations with $\mathcal{D}_j$ denoting the set of observations from the j -th population. We call the algorithm Perfect if $J = K$ and there exists a permutation π of $\{1, 2, \dots, K\}$ such that $C_j = \mathcal{D}_{\pi(j)}$ for each $j = 1, 2, \dots, K$.

Theorem 2: Consider a dataset $\mathcal{D}$ of n observations of sizes $n_1, n_2, \dots, n_K$ from $K > 1$ populations. Assume that

Algorithm: CURBS-II***Input:** Dataset $\mathcal{D}$ and specified number J (> 1) of clusters

- 1) Apply Algorithm BS* to $\mathcal{D}$.
- 2) If $J = 2$ then END, else set $i = 2$.
- 3) Apply Algorithm BS* to each of the i clusters obtained.
- 4) Compute all possible pairwise d_w^* -distances between the $2i$ clusters obtained in Step 3 and arrange them in the increasing order of magnitude.
- 5) Merge those $(i - 1)$ pair(s) of clusters obtained in Step 3 having the least d_w^* -distances.
- 6) If $i = J - 1$ END, else update $i = i + 1$ and go to Step 3.

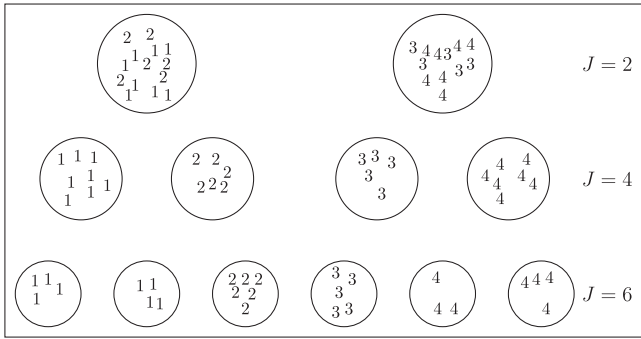
Output: The clusters $C_1, C_2, \dots, C_J$.

Fig. 2. A clustering algorithm which is POP at 4.

$\liminf_{n \rightarrow \infty} (n_i/n) > 0$ for all $i = 1, 2, \dots, K$. Then there exists $n^* \geq 1$ such that for any $n \geq n^*$, the Algorithm CURBS-I* is Perfect.

B. Oracle Version of Algorithm CURBS-II

The oracle version of the Algorithm CURBS-II, referred to as the Algorithm CURBS-II*, that uses the Algorithm BS* repeatedly along with some merging steps, are detailed above.

Algorithm CURBS-II* has an order preserving property in the oracle scenario. The concept of Perfect and Order Preserving (POP) clustering algorithm was introduced in [32].

Definition IV.11: For any fixed J , let $G_1^J, \dots, G_J^J$ be J clusters estimated using a clustering algorithm on $\mathcal{D} = \cup_{i=1}^K \mathcal{D}_i$, which consists of observations from K populations. We call the algorithm Perfect and Order Preserving (POP) at K if the following conditions hold.

- a) When $J = K$, the clustering algorithm is perfect.
- b) When $J < K$, then for every $i = 1, 2, \dots, K$, there exists a unique $l \leq J$ such that $\mathcal{D}_i \subseteq G_l^J$.
- c) When $J > K$, then for every $l = 1, 2, \dots, J$, there exists a unique $i \leq K$ such that $G_l^J \subseteq \mathcal{D}_i$.

Fig. 2 demonstrates the output of a POP clustering algorithm with $K = 4$ populations. We use only one value of $J < K$ and one value of $J > K$ in the figure for brevity. When $J = 2$, a POP algorithm will yield two clusters such that observations from any of the four populations (denoted by the 1's, 2's, 3's,

4's) will lie in exactly one of the two clusters. Thus, there is no sharing of observations from a single population between the two clusters. When $J = 4$, the algorithm will yield four clusters, each containing observations from a single population (perfect clustering). On the other hand, when $J = 6$, each cluster will contain observations from exactly one population. Hence, Fig. 2 demonstrates all three cases in Definition IV.11.

Theorem 3: The Algorithm CURBS-II* applied on a dataset $\mathcal{D}$ containing observations from K (≥ 2) populations is POP in the oracle scenario.

This theorem states that if the specified cluster number J is equal to the true number of populations K , then the Algorithm CURBS-II* provides perfect clustering. On the other hand, if $J < K$, then the Algorithm CURBS-II* produces clusters with the property that no two of them have observations from any given population. Finally, if $J > K$, then each of the clusters obtained will contain observations from a single population, and consequently, observations from a given population will be (possibly) split between more than one clusters (see Fig. 2 for an illustration).

V. NUMERICAL RESULTS

In this section, we investigate the behaviour of the Algorithms CURBS-I and CURBS-II. To compare the performance of Algorithm CURBS-I (denoted by C-I for brevity in the subsequent tables), we chose the following state-of-the-art clustering algorithms which have shown strong performance in the existing literature: FNN [24], FAE [25], FAEc [26], F1 and F2 [23], tf [20] and sf [21]. These algorithms either do not need any prior specification of the number of clusters (K) or need lower/upper bounds making them suitable for comparison with Algorithm CURBS-I. The codes for implementing FNN, FAE and FAEc were obtained from <https://github.com/samuveersingh/FAEclust>, the F1 and F2 methods are implemented using FADPclust package, the tf method using TFunHDDC package and the sf method using sasfunclust package in R. Further details on these algorithms are provided in the supplementary material. To compare the performance of Algorithm CURBS-II (denoted by C-II for brevity in the subsequent tables), we choose the following clustering algorithms which can be implemented by specifying the number of clusters (J): PC [10], kc [7] and D_H [10]. We also include the algorithms F1, F2 and tf considered earlier when they are implemented by setting the lower and upper bounds as J . The PC and the kc methods were implemented using the R package fdapace. The code for implementing the D_H procedure is available at <http://researchers.ms.unimelb.edu.au/~aurored/CodeClustering.zip>. To assess the effectiveness of clustering methods, we compute the average Rand index [33] by applying the clustering procedure to 100 independent datasets for each of the models detailed in the supplementary material. The Rand index measures the agreement between two partitions of the data and takes values between 0 and 1 with the latter value representing perfect association between the two partitions. In all the tables in this section, the maximum Rand index obtained in each row is shown in bold font and the second maximum

value is shown in italic font for ease of comparison. For some methods, there were instances when it did not converge or return any result. These cases are marked with a hyphen in the tables.

Some of these methods, particularly $\mathbf{tf}$ and $\mathbf{kC}$, have an enormous computational overload whereas the proposed algorithms are comparatively less computationally intensive. We report the average run times across various choices of sample sizes and number of clusters (equal sample sizes for all clusters) in Tables 4 and 5 of the supplementary material, which corroborate the polynomial time complexity discussed earlier for the proposed algorithms.

A. Analysis of Simulated Datasets

For the simulation study, we generated functional data from a variety of two-population and three-population models having mixtures of Gaussian processes (Model 1a - 6b and 8) and mixture of t processes (Models 7a and 7b). The specifications of each model are provided in the supplementary material along with other details on the implementation. Models 1a, 2a, 5a, 6a and 7a consider only difference in the mean functions of the component distributions, while the remaining models consider difference in the covariance operators. Since for functional data, difference in the covariance structure may mean either constant multiplicative factors in the eigenvalues of the component distributions (Models 1b, 2b, 5b, 6b and 7b), or it could mean that the system of eigenvalues or eigenfunctions differ across the component distributions (Models 3, 4 and 8). Differences in the rates of decay of the eigenvalues impart different degrees of smoothness to the component populations while changes in the eigenfunctions result in different structures of the component populations (see Fig. 1 in the supplementary material). These types of differences make clustering more difficult since the curves only differ either in their smoothness or in their shape with no overall changes in the mean. Models 7a and 7b are chosen to assess the performance of the competing procedures when clustering mixtures of heavy-tailed distributions, where the underlying stochastic processes have principal components following a Student's t distribution with 3 degrees of freedom. In Tables 1 and 2 of the supplementary material, we report the estimated pairwise MMD values between the component populations in each of the simulation models. Smaller MMD values indicate greater similarity between the corresponding populations and therefore a higher level of difficulty in reliably separating the associated observations into distinct clusters. To assess how the performance of the proposed algorithms varies with population separability, we plot the Rand index against the estimated MMD, as shown in Fig. 2 in the supplementary material. Algorithm CURBS-I exhibits a sharp performance jump once the MMD crosses a certain threshold, indicating its higher sensitivity, whereas Algorithm CURBS-II shows a smoother, more gradual improvement as the MMD increases.

Table I shows that C-I has the best overall performance among the competing procedures. It significantly outperforms all other methods for all the models differing in the covariance structure and even in some of the location-shift models with

TABLE I
RAND INDICES OF DIFFERENT CLUSTERING ALGORITHMS FOR UNSPECIFIED CLUSTER NUMBER

K	Sizes	Model	C-I	F1	F2	tf	sf	FAEc	FAE	FNN
2	(30,30)	1a	<i>0.98</i>	1	1	0.949	0.934	0.493	0.503	0.54
		1b	0.997	0.501	0.501	0.502	0.492	0.493	<i>0.508</i>	0.505
		2a	1	1	1	<i>0.995</i>	0.492	0.505	0.563	0.657
		2b	0.996	0.494	0.501	0.503	0.492	0.494	<i>0.51</i>	0.503
		3	0.639	0.492	0.506	0.494	0.492	0.494	<i>0.512</i>	0.504
		4	0.492	<i>0.718</i>	0.982	0.528	0.492	0.494	0.51	0.515
	(20,40)	1a	0.992	1	0.903	0.887	0.967	0.546	0.499	0.523
		1b	0.979	0.528	0.494	0.488	<i>0.548</i>	0.534	0.495	0.489
		2a	0.999	0.637	1	0.78	1	0.547	0.564	0.651
		2b	0.999	0.506	0.501	0.493	<i>0.548</i>	0.528	0.49	0.493
		3	0.976	0.528	0.548	0.492	<i>0.548</i>	<i>0.561</i>	0.509	0.498
		4	0.614	<i>0.892</i>	0.995	0.497	0.548	0.536	0.492	0.494
	(100,100)	1a	0.995	0.99	1	0.905	0.99	0.498	0.516	0.556
		1b	1	0.5	0.499	<i>0.501</i>	0.497	0.498	<i>0.501</i>	0.5
		2a	1	1	1	1	1	0.498	0.584	<i>0.613</i>
		2b	0.999	0.498	0.498	0.501	0.497	<i>0.528</i>	0.5	0.5
		3	0.999	0.498	0.498	0.498	0.497	0.498	<i>0.5</i>	0.5
		4	<i>0.847</i>	0.497	0.913	0.502	0.497	0.498	0.504	0.503
	(50,150)	1a	0.962	0.942	0.871	0.697	1	0.621	0.499	0.525
		1b	1	0.604	0.578	0.437	<i>0.623</i>	0.617	0.484	0.485
		2a	0.999	0.559	1	0.803	1	0.624	0.568	0.58
		2b	0.999	0.595	0.566	0.498	<i>0.623</i>	0.615	0.5	0.499
		3	0.997	0.509	0.498	0.498	<i>0.623</i>	<i>0.631</i>	0.503	0.489
		4	0.952	0.5	0.613	0.438	<i>0.623</i>	0.618	0.495	0.499
3	(30,30,30)	5a	0.998	1	1	0.632	0.326	0.378	0.585	0.634
		5b	0.985	0.385	0.411	<i>0.553</i>	0.326	0.359	0.519	0.538
		6a	0.991	0.775	0.775	<i>0.969</i>	0.775	0.346	0.633	0.644
		6b	0.998	0.4	0.416	<i>0.568</i>	0.326	0.343	0.508	0.53
		7a	0.972	0.775	1	0.638	0.721	0.373	0.614	0.638
		7b	0.969	0.375	0.359	<i>0.688</i>	0.326	0.349	0.555	0.498
	(20,30,40)	8	0.827	0.468	0.351	0.558	0.326	<i>0.597</i>	0.514	0.523
		5a	0.998	1	1	0.46	0.351	0.384	0.576	0.617
		5b	0.978	0.427	0.391	<i>0.546</i>	0.351	0.379	0.518	0.527
		6a	0.993	0.85	0.85	<i>0.909</i>	0.85	0.366	0.665	0.677
		6b	0.978	0.381	0.38	<i>0.581</i>	0.351	0.364	0.502	0.53
		7a	0.945	0.7	0.838	0.742	0.351	0.402	0.62	0.624
	(100,100,100)	7b	0.963	0.411	0.374	<i>0.639</i>	0.351	0.371	0.532	0.494
		8	0.844	0.41	0.435	0.548	0.351	<i>0.666</i>	0.509	0.527
		5a	0.998	1	1	0.686	0.331	0.335	0.686	0.627
		5b	0.991	0.477	0.34	<i>0.577</i>	0.331	0.335	0.536	0.518
		6a	0.997	0.777	0.777	0.694	0.774	0.336	0.657	0.629
		6b	0.999	0.346	0.398	0.495	0.331	0.335	<i>0.512</i>	0.505
	(150,25,25)	7a	0.974	0.775	1	0.854	0.725	0.335	0.648	0.646
		7b	0.98	0.353	0.379	<i>0.745</i>	0.717	0.335	0.584	0.512
		8	0.981	0.34	0.347	<i>0.58</i>	0.331	0.474	0.5	0.506
		5a	0.967	0.524	<i>0.766</i>	0.606	0.592	0.596	0.521	0.543
		5b	0.996	0.599	0.614	0.457	<i>0.639</i>	0.587	0.496	0.502
		6a	0.993	0.475	0.539	0.802	<i>0.812</i>	0.593	0.712	0.63
4	(30,30)	6b	0.978	0.568	0.541	0.462	0.592	0.588	0.5	0.5
		7a	0.949	0.635	0.993	0.897	0.918	0.586	0.608	0.65
		7b	0.989	0.518	0.581	0.583	0.885	0.589	0.556	0.519
		8	0.592	0.589	0.537	0.455	0.592	0.598	0.5	0.496

$K = 3$. None of the competing methods has a uniformly consistent performance and all of them fail in the models differing in the covariance operators (except F2 for small cluster sizes under Model 4). Performance degradation of C-I is observed only under Model 4 with small cluster sizes. This is due to the fact that Algorithm SCC yields inaccurate decisions. This issue diminishes as the cluster sizes increase. Fig. 3 shows a stacked bar plot of the estimated number of clusters using C-I and the top two performers (in terms of Rand index reported in Table I) in each of the three-population models under two sample size settings: (100,100,100) and (150,25,25). It is observed that the proposed algorithm perfectly estimates K under balanced sample sizes across all models. In the unbalanced setting, C-I is the only method that achieves near-perfect estimation for all models except Model 8.

We further evaluated Algorithm CURBS-I (C-I) in the single-population setting, where its behavior is governed by the Algorithm SCC. Using Models N1-N4 (see the supplementary material for details), we assessed the ability of the proposed algorithm to correctly return $\hat{K} = 1$. Table 3 in the supplementary material shows that C-I correctly identifies a single cluster in most cases.

We next present the performance of the Algorithm CURBS-II (C-II) and its competitors when the number of clusters is specified as J (see Table II). The results demonstrate that, in

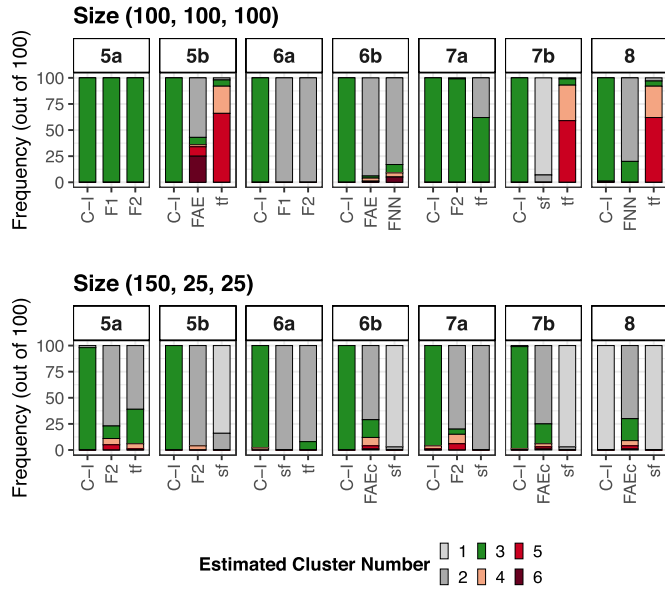


Fig. 3. Stacked bar plots of estimated number of clusters using C-I and the top two performers based on Rand index reported in Table I.

TABLE II
RAND INDICES OF DIFFERENT CLUSTERING METHODS FOR SPECIFIED CLUSTER NUMBER J WHICH IS CHOSEN TO BE EQUAL TO K

K	Sizes	Model	C-II	PC	kc	D _H	F1	F2	tf
2	(30,30)	1a	0.972	0.983	0.79	0.973	0.966	0.992	1
		1b	0.984	0.53	0.492	0.492	0.495	0.497	0.503
		2a	0.988	0.741	1	1	0.997	1	0.779
		2b	0.981	0.512	0.501	0.492	0.495	0.499	0.496
		3	0.992	0.53	0.496	0.52	0.495	0.497	0.5
		4	0.907	0.522	0.493	0.5	0.686	0.973	0.5
	(20,40)	1a	0.949	0.953	0.655	0.873	0.867	0.844	1
		1b	0.984	0.501	0.497	0.497	0.512	0.508	0.501
		2a	1	1	1	1	0.866	0.929	0.703
		2b	0.98	0.498	0.512	0.492	0.51	0.507	0.507
		3	0.977	0.498	0.494	0.456	0.532	0.532	0.508
		4	0.777	0.497	0.492	0.542	0.857	0.992	0.5
	(100,100)	1a	0.939	0.945	0.765	0.961	0.981	0.995	1
		1b	0.995	0.521	0.503	0.5	0.498	0.5	0.499
		2a	1	1	1	1	1	1	1
		2b	0.994	0.52	0.509	0.501	0.498	0.5	0.499
		3	0.964	0.503	0.496	0.464	0.499	0.5	0.5
		4	0.985	0.508	0.492	0.467	0.619	0.886	0.502
	(50,150)	1a	0.903	0.992	0.563	0.99	0.828	0.704	1
		1b	0.988	0.499	0.497	0.499	0.581	0.557	0.51
		2a	1	1	1	1	0.826	1	0.916
		2b	0.939	0.5	0.464	0.501	0.578	0.546	0.508
		3	0.997	0.5	0.498	0.521	0.578	0.567	0.509
		4	0.963	0.5	0.498	0.497	0.89	0.915	0.506
3	(30,30,30)	5a	0.979	0.973	1	0.675	1	1	–
		5b	0.982	0.464	0.557	0.555	0.456	0.451	0.555
		6a	0.98	0.896	0.771	0.779	0.94	0.996	1
		6b	0.978	0.479	0.548	0.589	0.466	0.454	0.553
		7a	0.98	0.947	1	1	0.996	0.998	0.985
		7b	0.98	0.41	0.641	0.555	0.471	0.464	0.681
		8	0.974	0.474	0.503	0.481	0.474	0.48	0.551
		–							
	(20,30,40)	5a	0.974	0.944	0.754	0.751	0.921	0.985	–
		5b	0.978	0.45	0.508	0.547	0.468	0.439	0.543
		6a	0.995	0.862	0.695	0.751	0.927	0.953	1
		6b	0.974	0.484	0.528	0.559	0.459	0.443	0.542
		7a	0.935	0.942	1	0.983	0.911	0.965	–
		7b	0.946	0.411	0.527	0.541	0.459	0.467	0.671
		8	0.806	0.468	0.495	0.492	0.486	0.489	0.542
		–							
	(100,100,100)	5a	0.991	0.98	1	0.781	1	1	0.736
		5b	0.986	0.473	0.564	0.546	0.413	0.437	0.545
		6a	0.964	0.922	0.726	0.78	0.979	0.997	1
		6b	0.994	0.49	0.557	0.586	0.416	0.45	0.545
		7a	0.987	0.948	0.991	0.996	0.999	0.996	0.858
		7b	0.955	0.436	0.554	0.584	0.422	0.452	0.734
		8	0.98	0.503	0.498	0.503	0.413	0.481	0.544
		–							
(150,25,25)	5a	0.99	0.786	1	0.616	0.595	0.681	0.611	
	5b	0.994	0.697	0.507	0.713	0.539	0.579	0.471	
	6a	0.963	0.981	0.969	0.626	0.612	0.602	1	
	6b	0.986	0.579	0.494	0.547	0.538	0.548	0.471	
	7a	0.9	0.922	1	1	0.632	0.775	0.701	
	7b	0.913	0.454	0.469	0.545	0.538	0.555	0.611	
	8	0.95	0.53	0.497	0.491	0.554	0.493	0.481	
	–								

the location-shift models, C-II does exceedingly well and provides consistent and near-perfect clustering performance, while none of its competitors performs uniformly well for all models. In the models with different covariance structures, all the competing algorithms have poor performance while C-II performs nearly accurately. Moreover, the performance of C-II improves upon the performance of C-I, which shows that it leverages on the additional information about the true number of clusters.

We additionally evaluated the proposed algorithms under three contamination schemes commonly encountered in functional data analysis, namely additive noise, isolated outliers, and missing fragments, in order to assess their performance relative to the competing methods considered in this study. In the additive noise setting, we perturbed the observed data from each of the previously described simulation models by adding i.i.d. Gaussian errors with mean zero and unit variance. To assess robustness to outliers, we randomly selected 10% of the observations from each component population and contaminated them using the isolated outlier mechanism described in [34]. This setup mimics situations in which a data-recording device experiences a brief malfunction, producing a localized spike while leaving the remainder of the functional trajectory unaffected. To examine the impact of missing parts over the functional domain, we randomly removed 10% of the observation grid points from each curve and applied the clustering algorithms to the resulting fragmented data. Additional details on each contamination mechanism are provided in the supplementary material.

Tables 6–11 in supplementary material collectively demonstrate the strong and consistent performance of C-I and C-II across noisy, outlier-contaminated, and fragmented data. For noisy data (Tables 6–7), C-I achieves near-perfect accuracy in models having difference in covariance, while C-II delivers uniformly superior performance, dominating even in highly unbalanced designs and the more difficult Models 3, 4, and 8. Under isolated outlier contamination (Tables 8–9), C-I remains highly accurate in models having difference in covariance operator but degrades under situations with extreme imbalance in the cluster sizes. On the other hand, C-II consistently achieves near-perfect clustering and outperforms all competitors, even in unbalanced settings. For fragmented functional data (Tables 10–11), C-I is robust in balanced to moderately unbalanced designs but deteriorates under severe imbalance, while C-II again attains near-perfect accuracy in models with difference in covariance and dominates in the difficult settings (Models 3 and 8). Overall, knowledge of the true number of clusters substantially enhances performance, with C-II emerging as the most reliable method across all data perturbations.

In Figs. 3, 4 and 5 in the supplementary material, we provide stacked bar plots (similar to Fig. 3 above) of the estimated number of clusters using C-I and the top two performers based on the Rand indices from Table 6 (noisy data), Table 8 (isolated outliers) and Table 10 (fragmented data) respectively in the supplementary material. It is observed that the overall conclusions remains the same as in Fig. 3 except in the unbalanced setting, where none of the methods correctly estimate K .

TABLE III

COMPARISON OF RAND INDICES ACROSS DIFFERENT DATASETS AND METHODS FOR UNSPECIFIED CLUSTER NUMBER

Dataset	K	C-I	F1	F2	tf	sf	FAEc	FAE	FNN
Phoneme	5	0.896	0.505	0.506	0.797	0.51	0.2	0.895	0.923
Meatspec	2	0.963	0.701	0.607	0.604	–	0.546	0.862	0.524
Berkeley	2	0.76	0.613	0.671	0.806	0.583	0.501	0.512	0.604
Flours	3	0.681	0.725	0.698	0.673	0.712	0.373	0.375	0.725
Wheat	2	0.609	0.519	0.495	0.835	–	0.515	0.729	0.593
Arrowhead	3	0.71	0.676	0.341	0.467	–	0.5	0.721	0.699
Satellite	2	0.732	0.534	0.524	0.689	0.616	0.5	0.774	0.71
Olive oil	4	0.89	0.686	–	0.689	–	0.367	0.759	0.765
BME	3	0.7	0.571	0.555	0.582	0.555	0.555	0.555	0.721
UMD	3	0.641	0.475	0.557	0.57	0.557	0.556	0.557	0.641
Meat	3	0.831	0.743	0.567	0.792	–	0.77	0.557	0.631
Synthetic Control	6	0.87	0.611	0.833	0.783	0.612	0.835	0.86	0.862

TABLE IV

COMPARISON OF RAND INDICES ACROSS DIFFERENT DATASETS AND METHODS FOR SPECIFIED CLUSTER NUMBER $J = K$

Dataset	K	C-II	PC	kc	D_H	F1	F2	tf
Phoneme	5	0.922	0.918	0.801	0.715	0.928	0.896	0.771
Meatspec	2	0.963	0.862	0.646	0.759	0.701	0.499	0.604
Berkeley	2	0.841	0.806	0.823	0.878	0.613	0.671	0.806
Flours	3	0.697	0.686	0.659	0.715	0.747	0.658	0.651
Wheat	2	0.728	0.728	–	0.715	0.519	0.505	0.835
Arrowhead	3	0.674	0.596	0.536	0.581	0.685	0.623	0.467
Satellite	2	0.732	0.653	0.635	0.635	0.534	0.524	0.499
Olive oil	4	0.89	0.845	–	0.718	0.783	–	–
BME	3	0.61	0.611	0.561	0.597	0.613	0.56	0.603
UMD	3	0.638	0.633	0.604	0.604	0.477	0.557	–
Meat	3	0.939	0.926	–	0.638	0.635	0.69	0.877
Synthetic Control	6	0.87	0.87	0.841	0.849	0.643	0.825	0.763

B. Analysis of Real Datasets

We applied the proposed clustering algorithms and the competing methods to twelve benchmark real datasets. Ten of these were obtained from the UEA and UCR Time Series Classification Repository, and the remaining two were the Phoneme dataset (from the speech-recognition domain) available in the `fds` package in R, and the Meatspectrum dataset (spectrometric measurements of fat content in meat) available in the `faraway` package. For clustering the Meatspectrum data, we have used the second order derivative curves since it is known from existing literature (see [35], [36] and [37]) that the spectra of a meat sample with high fat content often have two local maxima while a single maxima is mostly observed in case of low fat content. For the Berkeley Growth dataset and the Wheat dataset, clustering is performed using the first order derivative curves, which is consistent with prior works (see [38], [10]). We observe that C-I ranks among the top two methods in 10 of the 12 datasets when the number of clusters is unspecified and is the best performer in 5 datasets (see Table III). When the number of clusters is specified as the true number of populations, C-II ranks within the top two in 10 out of 12 datasets and achieves the best performance in 6 of them (see Table IV). No competing method exhibits such consistent performance in either setting, highlighting the robustness and practical usefulness of the proposed algorithms.

VI. EPILOGUE

Algorithm CURBS-I automatically learns the number of clusters from the data in addition to producing cluster assignments, and both Algorithms CURBS-I and CURBS-II successfully detect distributional differences between component populations, as supported by the oracle analysis. An important strength of our framework is its near-perfect performance when the component populations differ only in their covariance structure, a setting in which no existing method is known to perform comparably. Both methods adapt to functional data observed on irregular but dense grids via an initial smoothing step. The proposed algorithms apply directly to multivariate functional data and, more broadly,

to data in arbitrary metric spaces, since perfect-clustering guarantees require only that the kernel be characteristic on the underlying space. Since our algorithms use a kernel which involves L_2 distances between the data curves, they will be less efficient when the difference between the clusters are limited to only a small portion of the domain of the functional observations. Our methods are not yet adapted for functional data which are possibly observed over sparse grids, since curve-level smoothing in this setting produces high-variance reconstructions and unstable MMD estimates, which in turn reduce clustering accuracy. When the functional data contain missing fragments, especially large or multiple gaps in the domain, the performance of our methods decline. The issue is most pronounced when cluster separation stems from differences in covariance structure, because the smoothing needed to reconstruct incomplete curves can bias covariance estimates and thereby obscures the component-level differences. A potential remedy for sparse or fragmented designs is to project the data into an appropriate finite-dimensional space, such as via random projections, before applying the proposed algorithms. We leave such extensions for future investigation.

VII. PROOFS AND TECHNICAL DETAILS

All the proofs and technical details along with the details omitted in the Section V are given in the supplementary material.

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